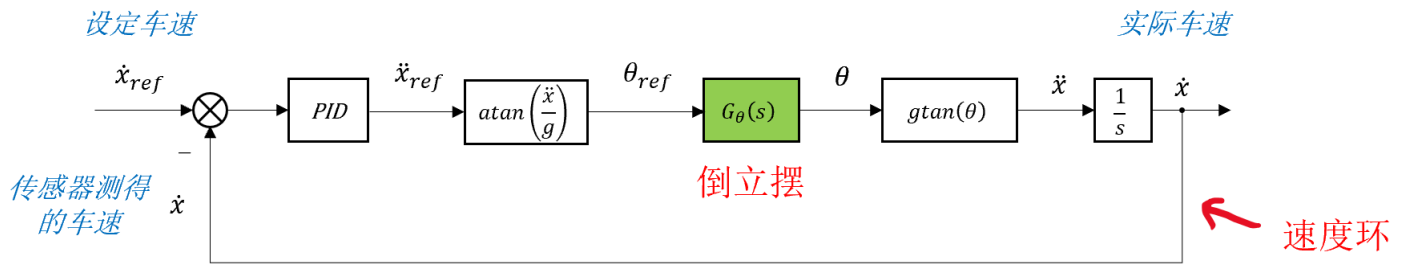


Speed Controller

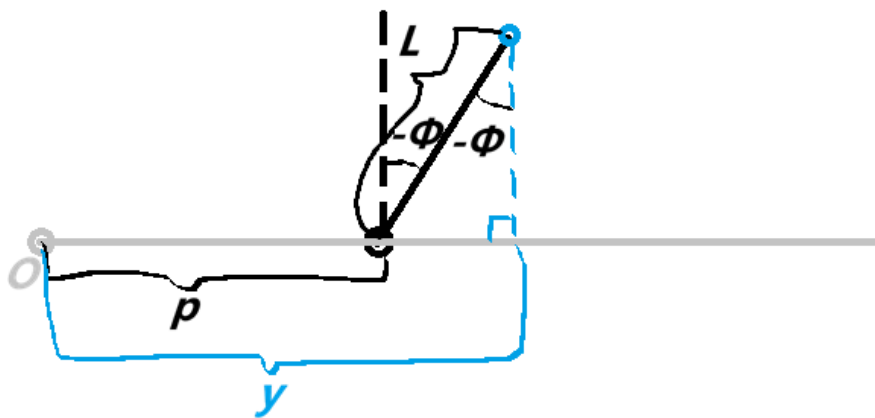
According to the ϕ controller, if there is a ϕ which is negative, a forward acceleration, i.e. $\ddot{y} > 0$, will be produced to maintain $\ddot{\phi} = 0$.

Therefore, speed could be controlled via controlling attitude.



e.g. I set $\dot{y}_{ref}$ as a positive number v . To form a negative ϕ , wheels have to rotate inversely. If $\dot{y}$ is simply calculated by ωR , which results $\dot{y}$ become smaller, $\dot{y}_{ref} - \dot{y}$ would become even larger due to wheels rotating inversely. That would ask $|\phi_{ref}|$ to become larger. Then wheels rotate more strongly in the inverse direction..... Eventually, a stable negative ϕ couldn't form. Instead, the car falls down!

In fact, if we only want to control the speed of the car body instead of the speed of wheels(i.e. define y as the position of the car body, not the position of the center of wheels. Meanwhile, $\dot{y}$ could be measured by MPU6050 fixed on the car body.), As a result, the positive feedback mentioned in the last paragraph would be turned into a negative feedback so that a stable speed can be produced. Detailed analysis is following:



From the geometrical relationship,

$$y = \dot{p} - L \sin \phi$$

$$\dot{y} = \ddot{p} - L \cos \phi \cdot \dot{\phi}$$

From angular momentum theorem,

$$mL^2 \dot{\phi} = (m\dot{p} \cos \phi + mg \sin \phi)L$$

$$\therefore L\dot{\phi} = \dot{p} \cos \phi + g \sin \phi$$

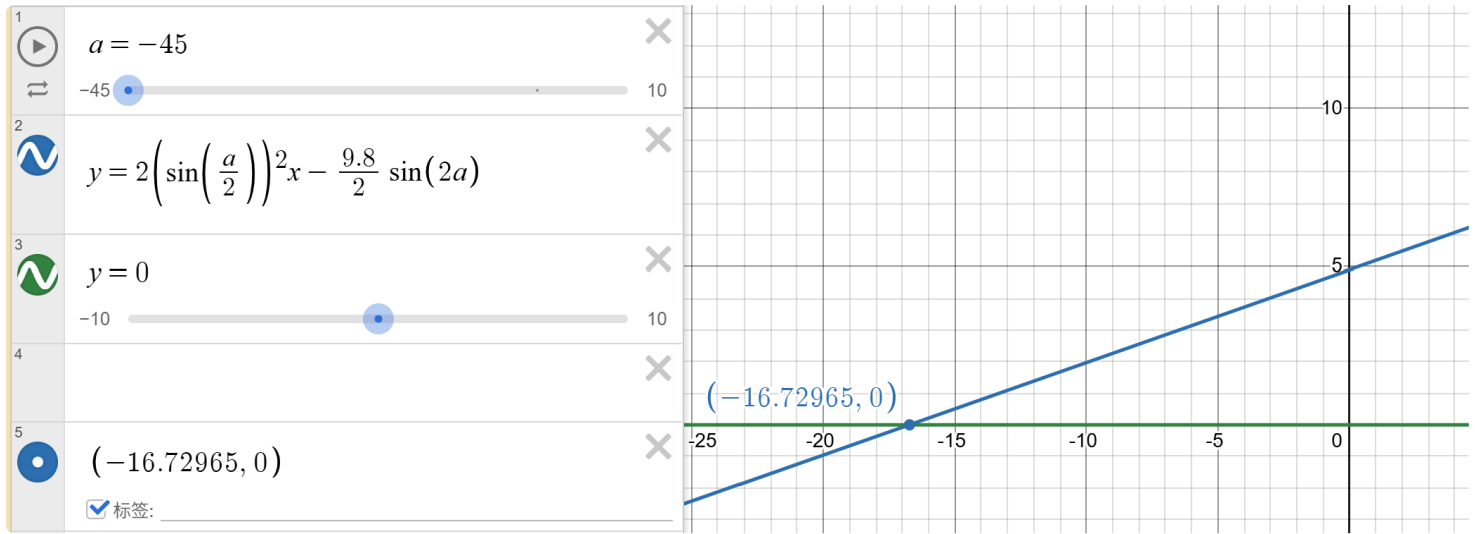
Combine two equations, we get:

$$\dot{y} = (1 - \cos^2 \phi)\dot{p} - g \sin \phi \cos \phi = 2 \sin^2 \frac{\phi}{2} \dot{p} - \frac{g}{2} \sin 2\phi \quad (\text{X})$$

In [倒立摆建模.pdf](#), we know that, when ϕ is stable, i.e. $\ddot{\phi} = 0$, $\ddot{y} = -g \tan \phi$.

Now just do some exaggerated assumption: $\ddot{y} = g$, so we have $\phi = -45^\circ$.

In this condition, let's check equation X, viewing $\dot{y}$ as a function of $\dot{p}$:



It shows that only when $\dot{p}$, which is generated by wheels rotating inversely, has an absolute larger than $16.73m/s^2$, will $\dot{y}$ become negative.

Actually for our balance car, $\ddot{y}$ must be much much smaller than g . Then the absolute of ϕ is much much smaller than 45° . In that case, the absolute of the zero of function X is much much larger than 16.73. Therefore, every time we command the car to move forward when $\dot{y} = 0$, (i.e. set a positive $\dot{y}_{ref}$, then the system need to create a negative ϕ , then wheels rotate inversely, which causes a negative $\dot{p}$) we are certain to get $\dot{y} > 0$. As a result, $\dot{y}_{ref} - y$ becomes smaller so that a negative feedback is formed, making a stable speed controller possible.